

Improved lower bound for the Gyarmati–Hennecart–Ruzsa sum-difference constant using masked digits and controlled carries

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Abstract

Zheng’s bounded-digit construction lower bounding the Gyarmati–Hennecart–Ruzsa sum-difference constant forms integers in a fixed base, restricts every digit to an interval from zero to a largest digit B , and imposes a bound on the sum of the digits. We first replace the complete digit interval by an arbitrary finite mask $M \subseteq \{0, \dots, B\}$. When the base is $2B + 1$, addition and subtraction are carry-free, and a method-of-types argument expresses the resulting limiting bound through two finite partition polynomials.

We next replace the carry-free base by an arbitrary integer $q > B$. On the difference side, each residue block is assigned its exact minimum witness cost; on the sum side, all representations of the same output block are combined before they are counted. Supermultiplicativity and submultiplicativity show that the corresponding block pressures have limits. A positive-vector certificate on a 351951-state accessible subgraph, together with a directed 8192-digit sum-pressure computation, proves

$$C_{3a} > 1.19519192.$$

The construction uses $B = 26972$, $q = 27022$, and $M = \langle 1971, 2016, 2100, 2628, 2688, 2800 \rangle \cap [0, B]$. Finally, we give a single explicitly specified finite set certifying $C_{3a} > 1.19102809$. This finite certificate uses neither a method-of-types limit nor an infinite-block pressure.

Formal verification and AI-use disclosure

An accompanying Lean 4/Mathlib formalization was produced with Aristotle (Harmonic) [14] and is included in the repository. It machine-checks the definition of C_{3a} , the Gyarmati–Hennecart–Ruzsa finite-set principle used here, the masked-digit construction, the full controlled finite-block bound, the controlled difference and sum pressure limits, the finite-state comparison arguments, the semigroup lemmas, and, in particular, the two conclusions $C_{3a} > 1.19519192$ and, from the explicit finite construction, $C_{3a} > 1.19102809$.

The source contains no `sorry`. Its large exact computations are evaluated with `native_decide`; consequently, the corresponding axiom report includes `Lean.trustCompiler` in addition to the standard Mathlib quotient, choice, and extensionality axioms.

The formal proof of Proposition 3 retains the possible nonzero self-inverse residue and implements the positional orientation described below by pairing consecutive occurrences from left to right. Its block-selection argument uses product-weight concentration with an even-parity restriction, rather than the fixed exact type displayed in the paper; it proves the same inequality. The formal numerical proof verifies the two pressure inequalities through exact truncated cost automata with 64003 difference states and 120001 sum states. These are sound finite relaxations of the exact

pressures, but they are not a line-by-line replay of the independent 351951-state and 8192-digit computations described in Section 4.3.

During this work, the author used several large language models (LLMs). OpenAI’s GPT-5.6 Sol model was used extensively in software development, analysis of search results, drafting and revising this paper, and preparation of the verification package. Google’s Gemini 3.1 Pro Preview was used in software development and analysis of search results; Google’s Gemini 3.5 Flash and Gemini 3.6 Flash were used to review the paper; and Anthropic’s Sonnet 5 was used in analysis of search results. Aristotle produced the Lean formalization, which was subsequently audited against the mathematical statements in this paper. Substantial portions of the initial prose and source code were generated or revised with this assistance. The author set the research goals, directed the computational work, selected which suggestions to pursue, ran the search and verification software, reviewed the mathematical arguments and outputs, and takes responsibility for all claims and references.

1 Introduction and the finite-set principle

Let C_{3a} be the supremum of the exponents θ with the following property: for every $K > 1$, there is a constant $c(K) > 0$ and there are pairs of nonempty finite sets $X, Y \subset \mathbb{Z}$, for arbitrarily large values of $|X|$, such that

$$|X + Y| \leq K|X|, \quad |X - Y| \geq c(K)|X + Y|^\theta.$$

Gyarmati, Hennecart, and Ruzsa proved the following finite-set lower-bound principle [3, Lemma in Section 2]: if $U \subset \mathbb{N}_0$ is finite and contains zero, and if

$$|U - U| < 2 \max U + 1,$$

then

$$C_{3a} \geq 1 + \frac{\log(|U - U|/|U + U|)}{\log(2 \max U + 1)}. \quad (1)$$

For any finite $U \subset \mathbb{N}_0$ containing zero, the elementary containment $U - U \subseteq [-\max U, \max U]$ gives only $|U - U| \leq 2 \max U + 1$, whereas the hypothesis of (1) is strict. Dilation supplies strictness without requiring any further information about U . If $\max U > 0$ and $V = 2U$, then

$$|V - V| = |U - U|, \quad |V + V| = |U + U|, \quad \max V = 2 \max U.$$

Consequently,

$$|V - V| = |U - U| \leq 2 \max U + 1 < 4 \max U + 1 = 2 \max V + 1.$$

Thus V satisfies the strict hypothesis of (1), whether or not U does, and

$$C_{3a} \geq 1 + \frac{\log(|U - U|/|U + U|)}{\log(4 \max U + 1)}. \quad (2)$$

The price is the replacement of $\log(2 \max U + 1)$ by $\log(4 \max U + 1)$. In the limiting constructions this does not change the exponential rate; in the finite certificate it adds less than $\log 2$ to a denominator of order 10^9 . We use the dilation because strictness for the undilated sets is not otherwise established.

Recent computational work on this constant includes AlphaEvolve [4], Gerbicz’s large finite construction [5], and Zheng’s large-deviation construction [6]. The *Optimization Constants in Mathematics* repository [2] subsequently listed the finite-certificate bounds 1.1740744 and 1.1835129324 [7, 8]. Two submissions merged in July 2026 then gave the limiting values 1.187326127925948 [9] and 1.19102809 [10]. The controlled-carry generalization proved here raises the limiting bound to 1.19519192, while the separate finite certificate gives 1.19102809 without asymptotic analysis.

Gerbicz introduced bounded base digits into the original construction, and Zheng analyzed the limiting exponential growth of a bounded-digit family. In Zheng’s version the allowed digit alphabet is the complete interval $\{0, \dots, B\}$. We replace that interval by a mask M . This can remove many possible sum digits while preserving difference digits d that admit witnesses $a, b \in M$ with $a - b = d$ and small $a + b$, so that they use little of the two digit-sum budgets in the construction. The multinomial counting estimate used below is standard; Proposition 2 is the construction-specific arbitrary-mask consequence. We first retain the carry-free base $2B + 1$ and then replace it by any $q > B$. The latter step allows controlled carries and leads to a transfer operator whose pressure is the $k \rightarrow \infty$ block limit. After proving and certifying the two bounds, we give a separate finite-depth witness and then discuss the numerical-semigroup structure that guided the search.

Theorem 1 (Main theorem). *The Gyarmati–Hennecart–Ruzsa constant satisfies*

$$C_{3a} > 1.19519192.$$

In addition, a single explicitly specified finite set, without recourse to either asymptotic theorem in this paper, certifies $C_{3a} > 1.19102809$.

The two assertions are proved in Sections 4.3 and 5, respectively.

2 The masked-digit bound

Let $M \subset \mathbb{N}_0$ be finite with $0 \in M$ and $B = \max M \geq 1$. Let

$$Q = 2B + 1, \quad S = M + M, \quad D = M - M.$$

For each $d \in D$, define its minimum witness cost

$$\kappa(d) = \min\{a + b : a, b \in M, a - b = d\}. \quad (3)$$

The function κ is even: $\kappa(-d) = \kappa(d)$. Define

$$P_+(x) = \sum_{s \in S} x^s, \quad P_-(x) = \sum_{d \in D} x^{\kappa(d)}. \quad (4)$$

We use the following standard lower bound for a multinomial coefficient; see, for example, Csiszár [1, Lemmas II.1 and II.2]. If E is a nonempty finite set and nonnegative integers m_e satisfy $\sum_{e \in E} m_e = m$, let $\pi_e = m_e/m$. Then

$$\log \frac{m!}{\prod_{e \in E} m_e!} \geq -m \sum_{e \in E} \pi_e \log \pi_e - \log \binom{m + |E| - 1}{|E| - 1}, \quad (5)$$

where $\log$ is the natural logarithm and $0 \log 0$ is interpreted as zero. For fixed E , the binomial coefficient in (5) is a polynomial in m of degree $|E| - 1$; equivalently,

$$\log \binom{m + |E| - 1}{|E| - 1} = O_E(\log m) = o(m).$$

Proposition 2 (Masked-digit bound). *For every $x \in (0, 1)$,*

$$C_{3a} \geq 1 + \frac{\log P_-(x) - \log P_+(x)}{\log(2B+1)}. \quad (6)$$

Consequently, one may take the supremum of the right-hand side over $x \in (0, 1)$.

Proof. Fix $x \in (0, 1)$ and define

$$p_d = \frac{x^{\kappa(d)}}{P_-(x)} \quad (d \in D). \quad (7)$$

Since κ is even, these weights satisfy $p_d = p_{-d}$. Let

$$\mu = \sum_{d \in D} p_d \kappa(d).$$

The entropy of p is

$$G(p) = - \sum_{d \in D} p_d \log p_d = \log P_-(x) - \mu \log x.$$

For each $d \in D$, choose $\alpha_d, \beta_d \in M$ such that

$$\alpha_d - \beta_d = d, \quad \alpha_d + \beta_d = \kappa(d),$$

with $(\alpha_{-d}, \beta_{-d}) = (\beta_d, \alpha_d)$; for $d = 0$, take $(\alpha_0, \beta_0) = (0, 0)$.

Let

$$R = \log \frac{P_-(x)}{P_+(x)}.$$

If $R \leq 0$, then (6) follows from the elementary bound $C_{3a} \geq 1$. For example, fix $0 < \delta < K - 1$ and take $X_n = \{0, \dots, n - 1\}$ and $Y_n = \{0, \dots, \lfloor \delta n \rfloor - 1\}$. For all sufficiently large n , these intervals satisfy $|X_n + Y_n| \leq K|X_n|$, and their sumset and difference set have the same cardinality. Assume henceforth that $R > 0$, and choose

$$0 < \varepsilon < \frac{R}{-\log x}.$$

For each positive integer m and each positive $d \in D$, define

$$n_d^{(m)} = n_{-d}^{(m)} = \lfloor mp_d \rfloor, \quad n_0^{(m)} = m - 2 \sum_{\substack{d \in D \\ d > 0}} \lfloor mp_d \rfloor.$$

These are nonnegative integers for every m : indeed, symmetry gives $2 \sum_{d > 0} p_d = 1 - p_0$, and hence

$$n_0^{(m)} \geq m - 2m \sum_{d > 0} p_d = mp_0 > 0.$$

Their sum is m , and, as $m \rightarrow \infty$, $n_d^{(m)}/m \rightarrow p_d$ for every $d \in D$. Write $q_d^{(m)} = n_d^{(m)}/m$. The number of sequences containing exactly $n_d^{(m)}$ copies of each d is the multinomial coefficient

$$\frac{m!}{\prod_{d \in D} n_d^{(m)}!}.$$

By (5), its logarithm is at least

$$-m \sum_{d \in D} q_d^{(m)} \log q_d^{(m)} - \log \binom{m + |D| - 1}{|D| - 1} = mG(p) + o(m).$$

Indeed, $q_d^{(m)} \rightarrow p_d$ for every d , the function $t \mapsto -t \log t$ is continuous on $[0, 1]$ when its value at zero is taken to be zero, and the final logarithmic term is $o(m)$ because D is fixed. Also, for all sufficiently large m ,

$$\sum_{d \in D} n_d^{(m)} \kappa(d) \leq (\mu + \varepsilon)m.$$

More explicitly, each type count differs from mp_d by $O_D(1)$, so the rounding changes the total cost by $O_{D,\kappa}(1)$ and the normalized entropy by $o(1)$. Set

$$L_m = \left\lfloor \frac{(\mu + \varepsilon)m}{2} \right\rfloor$$

and define

$$U_m = \left\{ \sum_{i=0}^{m-1} a_i Q^i : a_i \in M, \sum_{i=0}^{m-1} a_i \leq L_m \right\}. \quad (8)$$

Take any sequence $(d_0, \dots, d_{m-1})$ containing exactly $n_d^{(m)}$ copies of each $d \in D$. Symmetry of the counts gives $\sum_i d_i = 0$. Hence

$$\sum_i \alpha_{d_i} - \sum_i \beta_{d_i} = 0, \quad \sum_i \alpha_{d_i} + \sum_i \beta_{d_i} = \sum_i \kappa(d_i),$$

so each of the first two sums equals one half of the last. Each is an integer not exceeding $(\mu + \varepsilon)m/2$, and hence is at most L_m . Therefore

$$u = \sum_{i=0}^{m-1} \alpha_{d_i} Q^i, \quad v = \sum_{i=0}^{m-1} \beta_{d_i} Q^i$$

belong to U_m , and

$$u - v = \sum_{i=0}^{m-1} d_i Q^i.$$

The map from difference-digit sequences to integers is injective. Indeed, if

$$\sum_i d_i Q^i = \sum_i d'_i Q^i,$$

then, since $D \subseteq [-B, B]$, with $\delta_i = d_i - d'_i$ we have $\sum_i \delta_i Q^i = 0$ and $|\delta_i| \leq 2B < Q$. Reduction modulo $Q = 2B + 1$ therefore gives $\delta_0 = 0$; after division by Q , induction gives $\delta_i = 0$ for every i . The multinomial estimate above therefore yields

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log |U_m - U_m| \geq G(p). \quad (9)$$

Every element of $U_m + U_m$ has a carry-free base- Q digit sequence $(s_0, \dots, s_{m-1}) \in S^m$ with $\sum_i s_i \leq 2L_m$. If N_m is the number of all such digit sequences $(s_0, \dots, s_{m-1}) \in S^m$ satisfying $\sum_i s_i \leq 2L_m$, then, because $0 < x < 1$,

$$N_m x^{2L_m} \leq \sum_{(s_i) \in S^m} x^{\sum_i s_i} = P_+(x)^m.$$

Thus

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \log |U_m + U_m| \leq \log P_+(x) - (\mu + \varepsilon) \log x. \quad (10)$$

Combining (9), the identity defining $G(p)$, and (10) gives

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log \frac{|U_m - U_m|}{|U_m + U_m|} \geq R + \varepsilon \log x > 0. \quad (11)$$

Every $u \in U_m$ has digits at most B , so

$$u \leq B \sum_{i=0}^{m-1} Q^i = \frac{Q^m - 1}{2}.$$

On the other hand, $\mu > 0$ because $B \in D$ and $p_B > 0$, so $L_m \geq B$ for all sufficiently large m ; then $BQ^{m-1} \in U_m$. Consequently

$$2(Q-1)Q^{m-1} + 1 \leq 4 \max U_m + 1 \leq 2Q^m - 1$$

for all sufficiently large m , and hence

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log(4 \max U_m + 1) = \log Q.$$

The numerator in (2) is positive for all sufficiently large m by (11). Apply (2) to U_m and take the lower limit of the resulting right-hand side. The preceding denominator limit and (11) give

$$C_{3a} \geq 1 + \frac{R + \varepsilon \log x}{\log Q}.$$

For each fixed admissible ε , this conclusion follows after the construction-size limit $m \rightarrow \infty$. Since the left side is independent of ε , it is at least the supremum of these lower bounds. Letting $\varepsilon \downarrow 0$ in that supremum proves (6); no interchange of two limits is being used. $\square$

Remark 1 (Recovery of Zheng's optimization). *For the complete mask $M = \{0, 1, \dots, B\}$,*

$$P_+(x) = 1 + x + \dots + x^{2B}, \quad P_-(x) = 1 + 2x + 2x^2 + \dots + 2x^B.$$

Maximizing Proposition 2 for $B = 5$ reproduces Zheng's value 1.173077279785... [6]. Thus the proposition is a direct mask generalization of Zheng's limiting construction, although the proof above is self-contained apart from the finite-set principle (1) and the standard multinomial estimate (5).

Remark 2 (The balancing condition at an interior optimum). *Parameterize $x = e^{-\lambda}$, with $\lambda > 0$, and write*

$$F(\lambda) = \log P_-(e^{-\lambda}) - \log P_+(e^{-\lambda}).$$

Give S probability weights proportional to $e^{-\lambda s}$ and give D probability weights proportional to $e^{-\lambda \kappa(d)}$. Direct differentiation gives

$$F'(\lambda) = \mathbb{E}_+[s] - \mathbb{E}_-[\kappa(d)].$$

Consequently, every interior stationary point satisfies

$$\mathbb{E}_+[s] = \mathbb{E}_-[\kappa(d)].$$

This identity is useful when comparing masks: the optimizing parameter balances the typical weighted sum digit against the typical minimum difference-witness cost. It is not needed for the proof, which uses only one fixed value of λ .

3 A carry-free masked-digit example

Let

$$H = \langle 1518, 1524, 1587, 2024, 2032, 2116 \rangle \quad (12)$$

and let

$$M = H \cap [0, 17032].$$

Exact integer calculation gives

$$B = \max M = 17032, \quad Q = 2B + 1 = 34065,$$

and

$$|M| = 3121, \quad |M + M| = 18730, \quad |M - M| = 32369. \quad (13)$$

The conductor of H is 28922. Moreover,

$$[-13066, 13066] \cap \mathbb{Z} \subseteq M - M,$$

and the cheapest witness for difference one has cost

$$\kappa(1) = 14351.$$

The verification package contains the complete mask, complete sum support, and all values $(d, \kappa(d))$ for $d \in M - M$.

At the fixed decimal value

$$\lambda_0 = 0.000321149844434550835903084464712287774157626054669874186964, \quad x_0 = e^{-\lambda_0},$$

the partition sums have decimal expansions

$$P_-(x_0) = 668.3247325490738970446\dots, \quad P_+(x_0) = 91.03099370518693032667\dots,$$

so

$$\log \frac{P_-(x_0)}{P_+(x_0)} = 1.99357414328979926593\dots$$

and

$$1 + \frac{\log(P_-(x_0)/P_+(x_0))}{\log 34065} = 1.19102809750994590261\dots \quad (14)$$

The accompanying MPFR verifier [13] uses 384-bit arithmetic with directed rounding. It reconstructs all discrete data from the six generators and proves the conservative strict inequality

$$\boxed{C_{3a} > 1.19102809} \quad (\text{carry-free limit}). \quad (15)$$

No optimization claim is needed for the proof: Proposition 2 holds at the single fixed value λ_0 .

4 Controlled carries

We now modify the masked-digit construction by replacing the carry-free base $Q = 2B + 1$ by an arbitrary integer base $q > B$. The elements under consideration still have digits in $M \subseteq \{0, \dots, B\}$ and a bound on their total digit sum. Their digitwise sums have digits in $M + M$, as before, so the partition polynomial P_+ continues to give the required sumset upper bound. Signed difference digits lie in $D = M - M \subseteq [-B, B]$, but when $q < 2B + 1$ distinct signed digit words can determine the same integer after carrying. We must count distinct differences without counting such words more than once.

Fix a positive integer k and group the digit positions into consecutive blocks of k positions. Within one block, a signed digit vector $(d_0, \dots, d_{k-1}) \in D^k$ has integer value $\sum_{i=0}^{k-1} d_i q^i$. Two such values have the same residue modulo q^k precisely when they differ by a carry into the next block. We therefore retain, for each residue class, the least possible witness cost of a signed block with that residue. This removes all within-block carry collisions while preserving a large collection of differences. Formally, for $r \in \mathbb{Z}/q^k\mathbb{Z}$, define

$$c_k(r) = \min \left\{ \sum_{i=0}^{k-1} \kappa(d_i) : d_i \in D, \quad \sum_{i=0}^{k-1} d_i q^i \equiv r \pmod{q^k} \right\}, \quad (16)$$

with value $+\infty$ if no such digit vector exists. Thus a vector *represents r modulo q^k* when its integer value is congruent to r modulo q^k . Put

$$Z_k(x) = \sum_{r \in \mathbb{Z}/q^k\mathbb{Z}} x^{c_k(r)}, \quad x^{+\infty} := 0. \quad (17)$$

Proposition 3 (Controlled-carry block bound). *For every $k \geq 1$ and $x \in (0, 1)$,*

$$C_{3a} \geq 1 + \frac{k^{-1} \log Z_k(x) - \log P_+(x)}{\log q}. \quad (18)$$

Proof. If the right side is at most 1, use the elementary bound $C_{3a} \geq 1$. Otherwise fix k for the remainder of this proof, let

$$\mathcal{A}_k = \{r \in \mathbb{Z}/q^k\mathbb{Z} : c_k(r) < +\infty\}$$

be the attainable residue set, and set

$$R_k = \log Z_k(x) - k \log P_+(x) > 0, \quad p_r = \frac{x^{c_k(r)}}{Z_k(x)} \quad (r \in \mathcal{A}_k), \quad \mu = \sum_{r \in \mathcal{A}_k} p_r c_k(r).$$

Negation of all difference digits shows that $c_k(-r) = c_k(r)$ and hence $p_{-r} = p_r$. The entropy is

$$H(p) = \log Z_k(x) - \mu \log x. \quad (19)$$

Call a residue r *self-inverse* if $r = -r$ in $\mathbb{Z}/q^k\mathbb{Z}$, equivalently if $2r = 0$. The zero residue is always self-inverse; when q^k is even, the only other self-inverse residue is $h = q^k/2$. Let $\mathcal{R}_k$ contain one member of each two-element orbit $\{r, -r\} \subseteq \mathcal{A}_k$ under negation. For every $r \in \mathcal{R}_k$, choose a minimum-cost digit vector $d(r)$ representing r and set $d(-r) = -d(r)$. For zero use the all-zero vector. If h lies in $\mathcal{A}_k$, choose either minimum-cost vector $d(h)$; its negative has the same cost and also represents h , because $-h = h$ modulo q^k .

Let ℓ be the number of blocks in the construction. Define integer type counts explicitly by

$$\begin{aligned} N_r^{(\ell)} &= N_{-r}^{(\ell)} = \lfloor \ell p_r \rfloor & (r \in \mathcal{R}_k), \\ N_h^{(\ell)} &= 2 \left\lfloor \frac{\ell p_h}{2} \right\rfloor & \text{if } h \in \mathcal{A}_k, \\ N_0^{(\ell)} &= \ell - 2 \sum_{r \in \mathcal{R}_k} \lfloor \ell p_r \rfloor - N_h^{(\ell)}, \end{aligned}$$

where the terms involving h are omitted when $h \notin \mathcal{A}_k$. These counts sum to ℓ and are nonnegative. Indeed, every nonzero count was rounded downward, so the defining identity for the probabilities gives $N_0^{(\ell)} \geq \ell p_0 > 0$. Moreover,

$$\frac{N_r^{(\ell)}}{\ell} \longrightarrow p_r \quad (\ell \rightarrow \infty, r \in \mathcal{A}_k).$$

Since k is fixed, the alphabet $\mathcal{A}_k$ is finite. Consequently, the total rounding error in the cost is bounded by a constant depending on k (and on the fixed data M, q, x), and its contribution after division by ℓ tends to zero. The multinomial estimate (5) gives

$$\log \frac{\ell!}{\prod_{r \in \mathcal{A}_k} N_r^{(\ell)}!} = \ell H(p) + o_k(\ell), \quad (20)$$

and, for every fixed $\varepsilon > 0$ and all sufficiently large ℓ ,

$$\sum_{r \in \mathcal{A}_k} N_r^{(\ell)} c_k(r) \leq (\mu + \varepsilon)\ell.$$

When $h \in \mathcal{A}_k$, order its occurrences in each residue word from left to right, starting with index zero, and orient the even-indexed occurrences as $d(h)$ and the odd-indexed occurrences as $-d(h)$. Thus consecutive occurrences are paired with opposite orientations. The definition of $N_h^{(\ell)}$ makes the two counts equal. For every $r \in \mathcal{R}_k$, the equal counts and the convention $d(-r) = -d(r)$ cancel their signed digit vectors; the two orientations at h cancel in the same way. Consequently, the sum of all $k\ell$ unweighted difference digits is zero.

Use the symmetric minimum witnesses α_d, β_d fixed in the proof of Proposition 2. The total α -digit sum and total β -digit sum are then equal, and each is half the total cost. With

$$L_\ell = \left\lfloor \frac{(\mu + \varepsilon)\ell}{2} \right\rfloor,$$

let U_ℓ be the set of base- q integers with $k\ell$ digits in M and total digit sum at most L_ℓ . Every selected block-residue word supplies a difference of two members of U_ℓ .

For clarity, if a chosen signed block has integer value

$$R_j = \sum_{i=0}^{k-1} d_{jk+i} q^i,$$

then

$$|R_j| \leq B \frac{q^k - 1}{q - 1} \leq q^k - 1.$$

Hence $R_j \equiv r_j \pmod{q^k}$ and $0 \leq r_j < q^k$ imply $R_j \in \{r_j, r_j - q^k\}$. We now verify injectivity of the complete encoding. Put $Q = q^k$. Suppose two oriented residue words give the same integer

difference. Reduction modulo Q determines their first canonical residue, so the first residues agree. The corresponding signed blocks also agree as integers: for an ordinary residue the chosen block is fixed, while for h the sign is determined by the parity of the number of earlier occurrences of h . After subtracting this common first block and dividing by Q , the same argument applies to the two suffixes, with the same parity offset because the removed residues agree. Induction recovers the entire residue word. Thus distinct block-residue words give distinct integer differences.

The prefix dependence of the alternating rule is essential to this decoding argument. A rule assigning $d(h)$ to the first half of all occurrences and $-d(h)$ to the second half depends on the total number of occurrences in the whole word and need not be injective. The accompanying Lean development includes an explicit collision for that alternative rule. Thus

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log |U_\ell - U_\ell| \geq H(p).$$

Every member of $U_\ell + U_\ell$ is produced by some word in $S^{k\ell}$ of total at most $2L_\ell$. Uniqueness is unnecessary because these words are used only for an upper bound. Therefore

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell} \log |U_\ell + U_\ell| \leq k \log P_+(x) - (\mu + \varepsilon) \log x.$$

The logarithmic ratio consequently has lower rate $R_k + \varepsilon \log x$, which is positive if $0 < \varepsilon < R_k/(-\log x)$.

Moreover, $\mu > 0$: the residue B is attainable and has positive cost and positive p -weight. Hence $L_\ell \geq B$ eventually and $Bq^{k\ell-1} \in U_\ell$. Together with $\max U_\ell \leq B(q^{k\ell} - 1)/(q - 1)$, this gives

$$\lim_{\ell \rightarrow \infty} \frac{1}{\ell} \log(4 \max U_\ell + 1) = k \log q.$$

For each fixed admissible ε , apply (2) after taking ℓ sufficiently large, exactly as in Proposition 2. Taking the supremum of the resulting bounds over ε and then letting $\varepsilon \downarrow 0$ proves (18). $\square$

Lemma 4 (Infinite-block pressure). *For fixed $x \in (0, 1)$, the limit*

$$\mathcal{P}(x) = \lim_{k \rightarrow \infty} \frac{1}{k} \log Z_k(x) = \sup_{k \geq 1} \frac{1}{k} \log Z_k(x)$$

exists, and

$$C_{3a} \geq 1 + \frac{\mathcal{P}(x) - \log P_+(x)}{\log q}. \quad (21)$$

Proof. Concatenating minimum-cost representatives of residues modulo q^k and q^ℓ shows that

$$c_{k+\ell}(r + q^k s) \leq c_k(r) + c_\ell(s).$$

The map $(r, s) \mapsto r + q^k s$ is a bijection modulo $q^{k+\ell}$, so $Z_{k+\ell}(x) \geq Z_k(x)Z_\ell(x)$. Thus $a_k = \log Z_k(x)$, with $a_0 = 0$, is superadditive. For completeness, if $k = tm + r$ with $0 \leq r < m$, repeated superadditivity gives $a_k \geq ta_m + a_r$. Since the finitely many a_r are bounded below, first letting $k \rightarrow \infty$ and then taking the supremum over m proves

$$\lim_{k \rightarrow \infty} \frac{a_k}{k} = \sup_{m \geq 1} \frac{a_m}{m};$$

this is Fekete's lemma in the present setting.

The order of quantifiers is important. Proposition 3 first fixes k and then lets the number ℓ of blocks tend to infinity; its rounding constants are allowed to depend on k . The proposition therefore proves a valid lower bound for each individual integer k . We now take the supremum of this already-established family of bounds. No estimate uniform in k , and no interchange of the limits $\ell \rightarrow \infty$ and $k \rightarrow \infty$, is required. Using the displayed identity for $\mathcal{P}(x)$ gives (21). $\square$

4.1 Exact pressure for distinct sums

The factor $P_+(x)$ used above counts sum-digit words, even when several words produce the same integer after carrying. The following construction combines all such representations before counting them.

Let $\mathcal{H}_+ = \{0, 1\}$ be the possible carries. For a canonical output digit $r \in \{0, \dots, q-1\}$, a transition $i \rightarrow j$ is possible when

$$s + i = r + qj \quad \text{for some } s \in S = M + M.$$

Its cost is s . For a canonical word $\mathbf{r} = (r_0, \dots, r_{\ell-1})$ and $i, j \in \mathcal{H}_+$, let $c_\ell^+(i, \mathbf{r}, j)$ be the least total cost of a path from i to j emitting $\mathbf{r}$, with value $+\infty$ if no such path exists. Set

$$A_\ell^+(x)_{ij} = \sum_{\mathbf{r} \in \{0, \dots, q-1\}^\ell} x^{c_\ell^+(i, \mathbf{r}, j)}. \quad (22)$$

Let $W_\ell(x)$ be the sum, over the distinct integer outputs of ℓ sum digits, of x raised to the minimum cost of producing that output. A canonical output word together with its terminal carry determines the integer output uniquely, so the row sum of $A_\ell^+(x)$ starting with carry zero is $W_\ell(x)$. Starting with carry one translates every represented integer by one and leaves all input digits and costs unchanged. Hence both row sums are $W_\ell(x)$, and therefore

$$A_\ell^+(x)\mathbf{1} = W_\ell(x)\mathbf{1}, \quad \rho(A_\ell^+(x)) = W_\ell(x). \quad (23)$$

Proposition 5 (Two-sided carry pressure). *Define*

$$\mathcal{P}_-(x) = \mathcal{P}(x) = \sup_{k \geq 1} \frac{1}{k} \log Z_k(x)$$

and

$$\mathcal{P}_+(x) = \inf_{\ell \geq 1} \frac{1}{\ell} \log W_\ell(x).$$

Then the second infimum is a limit, and

$$C_{3a} \geq 1 + \frac{\mathcal{P}_-(x) - \mathcal{P}_+(x)}{\log q}. \quad (24)$$

More explicitly, every $k, \ell \geq 1$ gives

$$C_{3a} \geq 1 + \frac{k^{-1} \log Z_k(x) - \ell^{-1} \log W_\ell(x)}{\log q}. \quad (25)$$

If a finite accessible difference-state set F and a positive vector v satisfy $T_{x,F}v \geq cv$, where $T_{x,F}$ is the principal submatrix of the difference carry operator T_x defined in Section 4.2, and if $W_\ell(x)^{1/\ell} \leq p$, then

$$C_{3a} \geq 1 + \frac{\log(c/p)}{\log q}. \quad (26)$$

Finally, each finite sum block has the rigorous error estimate

$$0 \leq \frac{1}{\ell} \log W_\ell(x) - \mathcal{P}_+(x) \leq \frac{\log 2}{\ell}. \quad (27)$$

Thus $\mathcal{P}_+(x)$ is the exact asymptotic minimum-cost pressure of distinct sum outputs.

Proof. Split a canonical output word of length $r + s$ at the block boundary. For fixed boundary carries, the weight of a concatenated output is the maximum, over the two possible internal carries, of the two products of minimum-cost block weights. The corresponding entry of the ordinary matrix product $A_r^+(x)A_s^+(x)$ is the sum of those at most two products. Consequently, after summing over outputs and terminal carries,

$$W_{r+s}(x) \leq W_r(x)W_s(x) \leq 2W_{r+s}(x). \quad (28)$$

The left inequality makes $\log W_\ell(x)$ subadditive. Fekete's lemma and (23) give

$$\lim_{\ell \rightarrow \infty} \frac{1}{\ell} \log W_\ell(x) = \inf_{\ell \geq 1} \frac{1}{\ell} \log W_\ell(x) = \mathcal{P}_+(x).$$

Fix k and ℓ , and take the construction length N to be a common multiple of them. The difference argument in Proposition 3 supplies the same entropy lower bound per k -digit block. If both input digit sums are at most L , every element of the sumset has minimum sum-digit cost at most $2L$. Since $0 < x < 1$,

$$|U + U|x^{2L} \leq W_N(x) \leq W_\ell(x)^{N/\ell}.$$

The digit-budget term cancels the identical term in the difference entropy calculation. Dividing by N and then taking the construction-size limit proves (25). Taking the supremum over k and the infimum over ℓ proves (24).

Proposition 6 gives $\mathcal{P}_-(x) \geq \log c$, while $\mathcal{P}_+(x) \leq \ell^{-1} \log W_\ell(x) \leq \log p$; substituting these two inequalities in (24) proves (26). Finally, iterating the right inequality in (28) gives

$$W_{n\ell}(x) \geq 2^{-(n-1)} W_\ell(x)^n.$$

Taking logarithms, dividing by $n\ell$, and letting $n \rightarrow \infty$ proves the upper error bound in (27); the lower bound follows from the definition of the infimum. $\square$

4.2 The carry-state transfer operator

Put $w(d) = \kappa(d)$ for $d \in D$ and $w(d) = +\infty$ otherwise, and define for $0 \leq r \leq q$

$$A_r^+ = w(r), \quad A_r^- = w(q - r).$$

Thus A_r^+ is the cost of signed digit r , while A_r^- is the cost of signed digit $r - q$; the latter equality uses the evenness of w . Here $A_q^+ = +\infty$ and $A_q^- = w(0) = 0$. For a canonical prefix residue $R \in \{0, \dots, q^j - 1\}$ for which at least one signed word exists, define $C_j^{(0)}(R)$ and $C_j^{(1)}(R)$ to be the minimum costs of signed j -digit words whose integer values are, respectively, R and $R - q^j$; an impossible value has cost $+\infty$. These are the only two possible integer values with residue R , because

$$\left| \sum_{i < j} d_i q^i \right| \leq B \frac{q^j - 1}{q - 1} \leq q^j - 1.$$

Let $m_j(R) = \min(C_j^{(0)}(R), C_j^{(1)}(R))$ and normalize by setting

$$u_0 = C_j^{(0)}(R) - m_j(R), \quad u_1 = C_j^{(1)}(R) - m_j(R).$$

At least one of u_0, u_1 is zero. The state is

$$\delta = u_1 - u_0 \in \mathbb{Z} \cup \{-\infty, +\infty\},$$

with the conventions $\delta = +\infty$ when $u_1 = +\infty$ and $\delta = -\infty$ when $u_0 = +\infty$. Thus δ records both which of the two representatives is cheaper and their exact cost difference. For the empty prefix, $C_0^{(0)}(0) = 0$ and $C_0^{(1)}(0) = +\infty$, so the initial state is $+\infty$. Conversely, δ uniquely determines the normalized pair: it is $(0, \delta)$ when $0 \leq \delta < +\infty$, $(-\delta, 0)$ when $-\infty < \delta < 0$, $(0, +\infty)$ when $\delta = +\infty$, and $(+\infty, 0)$ when $\delta = -\infty$. Hence the update below depends only on the current state and the next canonical digit.

On appending the next canonical digit $r \in \{0, \dots, q-1\}$, set

$$v_0 = \min(u_0 + A_r^+, u_1 + A_{r+1}^+), \quad v_1 = \min(u_0 + A_r^-, u_1 + A_{r+1}^-). \quad (29)$$

If at least one value is finite, let

$$a(\delta, r) = \min(v_0, v_1), \quad \Phi(\delta, r) = v_1 - v_0$$

with the same extended convention. The definitions of A_q^+ and A_q^- make (29) valid also for $r = q-1$, including the borrow across the new boundary. Induction on the target digits shows that the sum of the increments a along a target word is exactly the minimum cost $c_k(R)$ of its terminal residue $R \in \mathbb{Z}/q^k\mathbb{Z}$.

Let T_x be the reachable-state kernel

$$T_x(\delta, \delta') = \sum_{\substack{0 \leq r < q \\ \Phi(\delta, r) = \delta'}} x^{a(\delta, r)}. \quad (30)$$

For vectors u and v indexed by the reachable states, write $u \cdot v = \sum_s u_s v_s$. Let $e_{+\infty}$ denote the coordinate vector supported at the initial state $+\infty$, and let $\mathbf{1}$ denote the vector whose coordinates are all one. Starting from $\delta = +\infty$, we then have the exact scalar path identity

$$Z_k(x) = e_{+\infty} \cdot (T_x^k \mathbf{1}). \quad (31)$$

Proposition 6 (Finite accessible-subgraph certificate). *Let $\mathcal{S}$ be the set of states reachable from $+\infty$. Suppose $F \subseteq \mathcal{S}$ is finite and there is an integer $h \geq 0$ such that*

$$(T_x^h)(+\infty, s) > 0 \quad (s \in F).$$

Let $T_{x,F} = (T_x(s, t))_{s,t \in F}$ be the principal submatrix indexed by F , and let $\rho(T_{x,F})$ denote its spectral radius, that is, the largest modulus of one of its complex eigenvalues. Then

$$\mathcal{P}(x) \geq \log \rho(T_{x,F}).$$

Here and below, $\log 0 = -\infty$ when the retained matrix is nilpotent. If $c > 0$, $v > 0$, and $T_{x,F} v \geq cv$ componentwise, then

$$C_{3a} \geq 1 + \frac{\log c - \log P_+(x)}{\log q}. \quad (32)$$

Proof. Define $b_s = (T_x^h)(+\infty, s) > 0$ for $s \in F$. In the path expansion of (31), retain only paths that reach a state of F after h steps and remain in F for the next n steps. Since every path weight is nonnegative,

$$Z_{h+n}(x) \geq b \cdot (T_{x,F}^n \mathbf{1}).$$

No irreducibility assumption is needed. Put $A = T_{x,F}$ and $b_* = \min_{s \in F} b_s > 0$. For the induced matrix 1-norm,

$$b \cdot (A^n \mathbf{1}) \geq b_* \mathbf{1} \cdot (A^n \mathbf{1}) \geq b_* \|A^n\|_1.$$

The Jordan normal form of the finite matrix A gives $\lim_{n \rightarrow \infty} \|A^n\|_1^{1/n} = \rho(A)$: each Jordan block with eigenvalue λ contributes a polynomial in n times $|\lambda|^n$. Hence the preceding path restriction and the existence of the pressure limit imply $\mathcal{P}(x) \geq \log \rho(A)$ (with the case $\rho(A) = 0$ trivial).

Finally, if $v > 0$ and $Av \geq cv$, then $A^n v \geq c^n v$ for every n . Writing $v_* = \min_s v_s$ and $v^* = \max_s v_s$ gives, for the induced infinity norm,

$$\|A^n\|_\infty \geq c^n \frac{v_*}{v^*}.$$

The same Jordan-form norm limit yields $\rho(A) \geq c$. This is the positive-vector form of the Collatz–Wielandt bound, proved here for the possibly reducible matrix A . Equation (32) now follows from Lemma 4. $\square$

4.3 The controlled-carry certificate

Take

$$B = 26972, \quad q = 27022, \quad M = \langle 1971, 2016, 2100, 2628, 2688, 2800 \rangle \cap [0, B], \quad (33)$$

and

$$x = 0x1.ffde827adc0fep - 1 = 0.9997444891905898 \dots \quad (34)$$

Exact reconstruction gives $|M| = 5869$ and $|M + M| = 31274$. For $j = 1, \dots, 8$, let

$$F_j = \{s \in \mathcal{S} : (T_x^j)(+\infty, s) > 0\}.$$

The verifier obtains

$$|F_1| = 7451, \quad |F_2| = 49009, \quad |F_3| = 109851, \quad |F_4| = 161577,$$

$$|F_5| = 210577, \quad |F_6| = 258233, \quad |F_7| = 305267, \quad |F_8| = 351951.$$

It checks a positive dyadic vector componentwise for T_{x, F_8} and proves

$$\rho(T_{x, F_8}) > 582.117820.$$

The file `certificate/controlled/pf8_vector.hex` contains the vector v , one coordinate for each state of F_8 . Each line gives the integer state label followed by the corresponding positive binary64 value in hexadecimal notation. The verifier reconstructs F_8 independently, checks that the labels occur in the required order, and then verifies the componentwise inequality

$$T_{x, F_8} v > 582.117820v.$$

All powers in the difference matrix are rounded downward. A row contains at most q multiplications and q additions; the verifier uses the deliberately larger relative-error allowance $4q2^{-52}$ in every Collatz inequality. Thus the matrix calculation is one-sided rather than a heuristic eigenvalue estimate.

On the sum side, directed dynamic programming constructs $W_{8192}(x)$ and proves

$$W_{8192}(x)^{1/8192} < 79.428331.$$

Every multiplication and addition in this calculation is rounded upward. The values are rescaled by exact powers of two to avoid overflow and underflow. The final comparison is made against

a downward-rounded power of the rational number $79428331/10^6$. By (27), replacing the exact infinite sum pressure by this finite block changes the resulting exponent by at most

$$\frac{\log 2}{8192 \log 27022} < 8.292 \times 10^{-6}.$$

This estimate is diagnostic only: the displayed finite-block upper bound is already rigorous and is what is used in the theorem.

It remains to compare the rational pressure bounds with the reported decimal. The accompanying Python script uses only exact rational arithmetic. It range-reduces each logarithm by powers of two and encloses $\log y = 2 \operatorname{atanh}((y-1)/(y+1))$ between a positive rational partial sum and an explicit geometric tail. It certifies

$$1 + \frac{\log(582.117820/79.428331)}{\log 27022} > 1.19519192.$$

Equations (26) and (33) therefore prove

$$\boxed{C_{3a} > 1.19519192}. \quad (35)$$

5 A direct finite-depth certificate

For a bound that does not invoke either limiting theorem, return to the carry-free mask (12), with $B = 17032$ and $Q = 34065$, and set

$$x_f = 0.9996789017186568, \quad N = 10^{13}.$$

Let

$$W = \sum_{d \in D} x_f^{\kappa(d)}, \quad n_d = \left\lfloor \frac{N x_f^{\kappa(d)}}{W} \right\rfloor \quad (d > 0), \quad n_{-d} = n_d,$$

and let $n_0 = N - 2 \sum_{d>0} n_d$. Define

$$L = \frac{1}{2} \sum_{d \in D} n_d \kappa(d) = 51500691976683641$$

and the explicit finite set

$$U = \left\{ \sum_{i=0}^{N-1} a_i Q^i : a_i \in M, \sum_i a_i \leq L \right\}.$$

The finite set to which (1) is applied is $V = 2U$, as in the dilation argument leading to (2). The elementary bound for U itself is only $|U - U| \leq 2 \max U + 1$; since strict inequality has not been established, applying (1) directly to U would leave a gap. Passing to V supplies the required strictness at the negligible cost of the additional $\log 2$ described in Section 1. For each d , use the symmetric minimum-cost witnesses (α_d, β_d) fixed in the proof of Proposition 2. There are exactly

$$D_N := \frac{N!}{n_0! \prod_{d>0} (n_d!)^2}$$

signed difference words $(d_0, \dots, d_{N-1})$ having the prescribed counts. Because $n_d = n_{-d}$, each such word satisfies $\sum_i d_i = 0$. Replacing each d_i by its fixed witness $\alpha_{d_i} - \beta_{d_i}$ gives two mask words.

Their digit sums are equal because $\sum_i (\alpha_{d_i} - \beta_{d_i}) = \sum_i d_i = 0$, and their sum is $\sum_i (\alpha_{d_i} + \beta_{d_i})$. Consequently, each digit sum is

$$\frac{1}{2} \sum_i (\alpha_{d_i} + \beta_{d_i}) = \frac{1}{2} \sum_{d \in D} n_d \kappa(d) = L.$$

They therefore encode two members of U , whose difference is $\sum_i d_i Q^i$. Finally, $d_i \in [-B, B]$ and $Q = 2B + 1$, so the carry-free injectivity established in Proposition 2 implies that two different signed words produce two different integers $\sum_i d_i Q^i$. Thus these D_N words give D_N distinct elements of $U - U$, and hence

$$|U - U| \geq D_N,$$

while the same elementary weighted count used earlier gives

$$|U + U| \leq S_N := P_+(x_f)^N x_f^{-2L}.$$

No limit is involved in either inequality. Since $\max U \leq (Q^N - 1)/2$, the dilated denominator satisfies

$$4 \max U + 1 \leq 2Q^N - 1 < 2Q^N.$$

Directed 512-bit MPFR evaluation [13] reconstructs all type counts (including $n_0 = 14962802398$) and certifies the strict inequalities

$$\begin{aligned} \log D_N &> 98126619915169.7944, \\ \log S_N &< 78190878820128.0915, \\ \log(2Q^N) &< 104360257432078.3785. \end{aligned}$$

Substitution in (2) gives a lower value $1.1910280942725\dots$ and proves the wholly finite statement

$$\boxed{C_{3a} > 1.19102809} \quad (\text{explicit finite depth } N = 10^{13}). \quad (36)$$

This value is weaker than both limiting bounds, as expected, but it can be recomputed directly without accepting the asymptotic arguments in Proposition 2 or Lemma 4.

6 Semigroup structure and computational motivation

This section explains the numerical-semigroup structure of the carry-free example in Section 3. It does not assert that a mask optimal for the carry-free base remains optimal after carries are allowed: the controlled-carry certificate uses the different mask

$$\langle 1971, 2016, 2100, 2628, 2688, 2800 \rangle \cap [0, 26972] = (\langle \{3, 4\} \times \{657, 672, 700\} \rangle) \cap [0, 26972].$$

Nevertheless, both successful masks arise from a 2×3 product grid, which suggests that the relation structure discussed below remains useful when the base is lowered even though the optimal parameters change.

A numerical semigroup is a cofinite additive submonoid of $\mathbb{N}_0$. Numerical semigroups are useful here because their missing elements suppress the low-degree part of P_+ , while additive relations among their generators create further sum collisions. The objective is nevertheless not determined by the Hilbert series or by the three cardinalities in (13). It also depends on the complete weighted minimum-cost function κ .

For a generating vector $G = (g_1, \dots, g_e)$, a useful untruncated analogue of κ is the residual cost function

$$\nu_G(d) = \min \left\{ \sum_i (z_i^+ + z_i^-) g_i : z_i^\pm \in \mathbb{N}_0, \quad \sum_i (z_i^+ - z_i^-) g_i = d \right\}.$$

Thus $\nu_G(d)$ is the least total cost of two semigroup elements whose difference is d . If a minimizing pair lies in $[0, B]$, then $\kappa(d) = \nu_G(d)$. Exact relations, with residual zero, help produce sum collisions; near relations with small nonzero residuals help produce many difference digits at low cost. The example combines both kinds.

6.1 A progression of carry-free constructions

The carry-free example emerged from a sequence of increasingly structured numerical-semigroup masks:

semigroup generators	B	resulting value
$\{24, 26, 36, 39\}$	189	1.1893936243...
$\{60, 63, 80, 84\}$	538	1.1902074763...
$\{312, 315, 336, 416, 420\}$	3084	1.1902381380...
$\{1518, 1524, 1587, 2024, 2032, 2116\}$	17032	1.1910280975...

The first two generator sets have the forms

$$12\langle 2, 3 \rangle + 13\langle 2, 3 \rangle \quad \text{and} \quad 20\langle 3, 4 \rangle + 21\langle 3, 4 \rangle.$$

In the general four-generator family $S\langle k, k+1 \rangle + (S+1)\langle k, k+1 \rangle$, the identity

$$Sk + (S+1)(k+1) = S(k+1) + (S+1)k + 1$$

produces a unit residual with shared support.

The former five-generator example is an iterated gluing:

$$\langle 312, 315, 336, 416, 420 \rangle = 21\langle 15, 16, 20 \rangle + 104\langle 3, 4 \rangle, \quad \langle 15, 16, 20 \rangle = 5\langle 3, 4 \rangle + 16\mathbb{N}_0.$$

Gluing is standard in the theory of complete-intersection numerical semigroups [11, 12]. Studying this decomposition led to searches parameterized by relations rather than by unrelated generator coordinates.

6.2 A balanced simple gluing in the columns

Write

$$C = \langle 506, 508, 529 \rangle.$$

The column semigroup has the exact simple-gluing description

$$C = 23\langle 22, 23 \rangle + 508\mathbb{N}_0, \quad 508 = 21 \cdot 22 + 2 \cdot 23. \quad (37)$$

More generally, for a simple gluing

$$C_{a,b,r,s} = a\langle r, s \rangle + b\mathbb{N}_0 = \langle ar, as, b \rangle,$$

where $\gcd(r, s) = \gcd(a, b) = 1$ and $b \in \langle r, s \rangle$, the two complete-intersection relation degrees are ars and ab , and its Hilbert series is

$$\sum_{n \in C_{a,b,r,s}} t^n = \frac{(1 - t^{ars})(1 - t^{ab})}{(1 - t^{ar})(1 - t^{as})(1 - t^b)}.$$

For these parameters

$$(r, s, a, b) = (22, 23, 23, 508),$$

the relation degrees are

$$ars = 11638, \quad ab = 11684,$$

differing by only 46, while the generators

$$ar = 506, \quad b = 508, \quad as = 529$$

are also clustered. The two balances have distinct meanings: $b \approx rs$ aligns the relation degrees, whereas $b \approx ar$ clusters two generators. Satisfying both naturally forces $a \approx s$; in the present example, $a = s$ exactly. Formula (37) also gives the column Frobenius number

$$23(22 \cdot 23 - 22 - 23) + 22 \cdot 508 = 21779.$$

6.3 The 2×3 generator grid

The semigroup is generated by the grid

$$\{3, 4\} \times \{506, 508, 529\} = \{1518, 1524, 1587, 2024, 2032, 2116\}. \quad (38)$$

For a column c , the two row generators have the exact relation

$$4(3c) = 3(4c).$$

For the three columns, these relations occur at degrees

$$6072, \quad 6096, \quad 6348,$$

well before the cutoff $B = 17032$.

The column relation $23 \cdot 506 = 22 \cdot 529$ also lifts to the grid:

$$5(1518) + 2(2024) = 2(1587) + 4(2116) = 11638.$$

The other relation in (37) can be lifted after adding $4 \cdot 529$ to both sides:

$$5(1524) + 2(2032) + 2116 = 7(1518) + 2(1587) = 13800.$$

Thus several exact relation layers are already active below the chosen cutoff, while the conductor of the full six-generator semigroup, 28922, remains substantially larger.

The same grid simultaneously gives cheap near relations. If $c_i < c_j$ are columns, then

$$(3c_i + 4c_j) - (4c_i + 3c_j) = c_j - c_i, \quad (39)$$

and the total cost of the two witnesses is $7(c_i + c_j)$. The column gaps are 2, 21, and 23, so the mask has the explicit bounds

$$\kappa(2) \leq 7098, \quad \kappa(21) \leq 7259, \quad \kappa(23) \leq 7245.$$

Because $\gcd(2, 21) = 1$, the cheap residual directions generate the full integer lattice, although combining them can increase witness cost and is subject to the cutoff. In contrast, the exact certificate gives $\kappa(1) = 14351$. This illustrates why minimizing $\kappa(1)$ alone is not an adequate design criterion: Proposition 2 uses the entire weighted distribution of $\kappa(d)$.

6.4 Structural properties suggested by the search

Searches around (38) found a narrow interior ridge: consecutive r, s , equality $a = s$, near equality $b \approx rs \approx ar$, and the row pair $\{3, 4\}$ were all favored. Relaxing the relation balance and adding a third binary product factor did not improve the best value found. These computations are evidence about the explored families, not an optimality proof.

The structural evidence suggests that a strong semigroup mask should have:

1. several exact relations becoming active below the cutoff;
2. several cheap near relations with small coprime residuals;
3. substantial sharing of generators among those relations;
4. clustered relation degrees, so that multiple collision mechanisms become active in the same range; and
5. delayed density, with the cutoff after the useful relation layers but before the conductor.

Product grids and gluings manufacture these features, but neither is logically required by Proposition 2. A natural broader program is therefore to prescribe a small weighted relation lattice first and solve the resulting Diophantine constraints for the generators. This includes low-rank grids with small circuit defects, carefully selected additive gluings, and constrained families described by numerical-semigroup coordinates. Such a relation-template approach is more closely tied to the objective than an unstructured search over generator tuples.

7 Verification and reproducibility

The source code, certificate files, verification scripts, and saved run logs are available in the repository at

<https://github.com/kleinwaks/masked-digit-sum-difference-bound>.

After this update package is integrated, the repository contains the following checks. Items 1–3 are the existing carry-free checks; items 4–6 are supplied or extended by this update, and item 7 is the accompanying formal verification.

1. `verification/verify_mpfr.cpp` reconstructs M , $M + M$, and $\kappa(d)$ using exact integer arithmetic and certifies (15) using MPFR directed rounding.
2. `verification/verify_python.py` independently reconstructs the exact discrete data using the Python standard library and checks the shipped certificate files and hashes.
3. `verification/verify_gp.gp` independently reconstructs the data in PARI/GP and evaluates the bound at high precision. This is a secondary numerical cross-check; the MPFR program supplies the rigorous floating-point certificate.
4. `verification/controlled/certify_candidate.cpp`, together with `pressure.hpp`, reconstructs the controlled-carry mask, all eight accessible state sets, the 351951 component Collatz certificate, and the directed 8192-digit sum-pressure bound.
5. `verification/controlled/verify_exponent.py` checks the final decimal exponent with exact rational logarithm enclosures. Its conclusion does not depend on a floating-point implementation of log.

6. `verification/verify_finite_mpfr.cpp` reconstructs the $N = 10^{13}$ finite type counts and certifies (36) with directed MPFR rounding. Its running time is governed by the fixed mask size, not by enumeration of the elements of U .
7. `formalization/lean/` contains the Lean 4/Mathlib formalization produced with Aristotle. Running its `verify.sh` builds the project, prints the principal theorem and axiom reports, and rejects any occurrence of `sorry`, `admit`, or an explicit project axiom.

References

- [1] I. Csizsár, *The method of types*, IEEE Transactions on Information Theory 44 (1998), no. 6, 2505–2523, <https://doi.org/10.1109/18.720546>, https://iest2.ie.cuhk.edu.hk/~whyung/post/manuscript/06_SAEF/method%20of%20types.pdf.
- [2] D. Davis, P. Ivanisvili, T. Tao, and contributors, *Optimization Constants in Mathematics*, GitHub repository, 2026, <https://github.com/teorth/optimizationproblems>.
- [3] K. Gyarmati, F. Hennecart, and I. Z. Ruzsa, *Sums and differences of finite sets*, Functiones et Approximatio Commentarii Mathematici 37 (2007), 175–186, <https://doi.org/10.7169/facm/1229618749>, <https://gyarmatikati.web.elte.hu/publ/sumdiffv.pdf>.
- [4] B. Georgiev, J. Gómez-Serrano, T. Tao, and A. Z. Wagner, *Mathematical exploration and discovery at scale*, arXiv:2511.02864, 2025, <https://arxiv.org/abs/2511.02864>.
- [5] R. Gerbicz, *Sums and differences of sets (improvement over AlphaEvolve)*, arXiv:2505.16105, 2025, <https://arxiv.org/abs/2505.16105>.
- [6] F. Zheng, *Sums and differences of sets: a further improvement over AlphaEvolve*, arXiv:2506.01896, 2025, <https://arxiv.org/abs/2506.01896>.
- [7] S. Griego, *Base-21 digit construction certificate for C_{3a}* , GitHub pull request #71 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/71>.
- [8] Mosaic Intelligence, *Exact-count certificate for problem 3a*, GitHub pull request #95 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/95>.
- [9] Numaro, *Certified lower bound $C_{3a} \geq 1.187326127925948$* , GitHub pull request #131 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/131>.
- [10] L. Kleinwaks, *Masked-semigroup digit bound $C_{3a} > 1.19102809$* , GitHub pull request #134 to the Optimization Constants repository, 2026, <https://github.com/teorth/optimizationproblems/pull/134>.
- [11] A. Assi and P. A. García-Sánchez, *Constructing the set of complete intersection numerical semigroups with a given Frobenius number*, arXiv:1204.4258, 2013, <https://arxiv.org/abs/1204.4258>.

- [12] A. Assi, P. A. García-Sánchez, and I. Ojeda, *Frobenius vectors, Hilbert series and gluings of affine semigroups*, Journal of Commutative Algebra 7 (2015), no. 3, 317–335, <https://doi.org/10.1216/JCA-2015-7-3-317>, <https://arxiv.org/abs/1311.1988>.
- [13] L. Fousse, G. Hanrot, V. Lefèvre, P. Pélicier, and P. Zimmermann, *MPFR: A multiple-precision binary floating-point library with correct rounding*, ACM Transactions on Mathematical Software 33 (2007), no. 2, Article 13, <https://doi.org/10.1145/1236463.1236468>, <https://inria.hal.science/inria-00070266v1/document>.
- [14] T. Achim et al., *Aristotle: IMO-level Automated Theorem Proving*, arXiv:2510.01346, 2025, <https://arxiv.org/abs/2510.01346>.